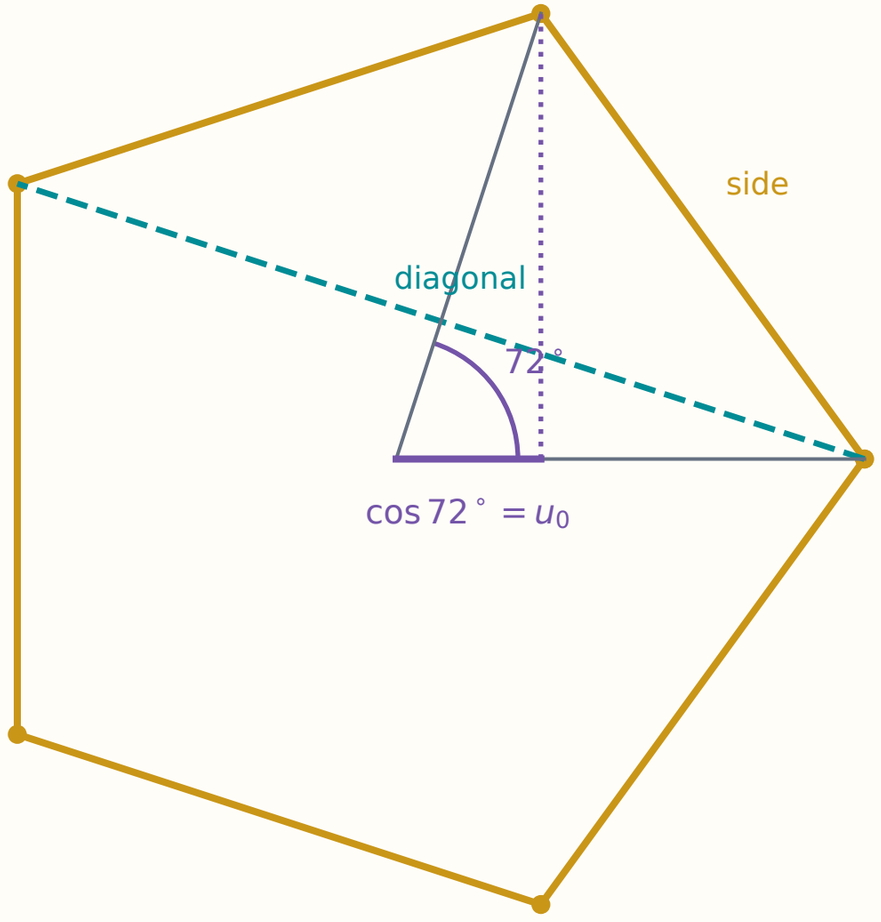
Unit pentagon: vertices 0, 72, 144, ... degrees



actual length ratio

side / diagonal $= \phi^{-1} = \gamma$

squared ratio $= \phi^{-2} = c_0$

On this unit circle, the horizontal projection has value $u_0 = \cos 72^\circ$.